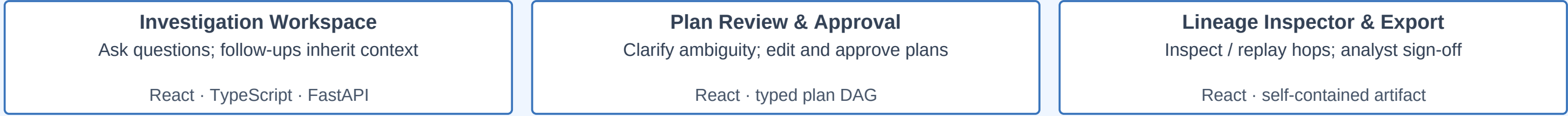


InsightPilot — System Architecture (Function + Technology)

Proposed design · Layered, agent-orchestrated architecture · CMPE 295A + 295B scope

CUSTOMER ENVIRONMENT — ONE ORGANISATION · SELF-HOSTED / DOCKER COMPOSE

PRESENTATION TIER



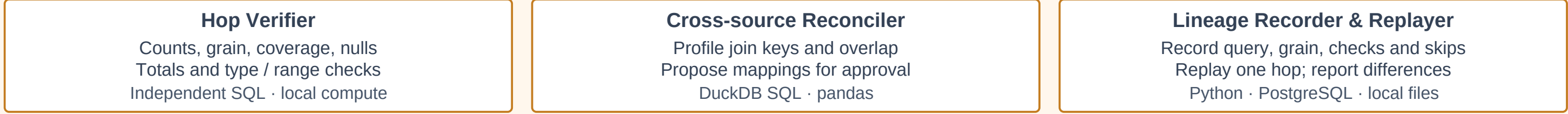
ORCHESTRATION TIER — CUSTOM PYTHON 3.12



SPECIALIST AGENT TIER — ALL TASKS AND RESULTS ROUTED THROUGH ORCHESTRATOR



VERIFICATION & LINEAGE TIER — CHECKS USE SQL / COMPUTE, NEVER LLM SELF-CRITIQUE



INTEGRATION TIER — READ-ONLY SOURCES; SEMANTIC DEFINITIONS HAVE SEPARATE PRECEDENCE



DATA, AUDIT & EVALUATION TIER



HTTPS

EXTERNAL MODEL PROVIDER

Frontier: planning / SQL
Small model: prose / chart choice
Customer's own API key
Question + schema; optional samples
PII-filtered / schema-only mode

Outside customer environment

Returned plan / SQL / prose
Data-source access stays local

KEY: Solid box = 295A scope; dashed box + [295B/later] = future scope. Solid arrow = requests/data; dashed arrow = control/response.

AGENT CONTROL: No direct agent-to-agent calls. Failed checks halt downstream work; 295B adds capped replanning / partial-result exit.

READING: Arrows summarise tier interactions, not execution order. Analyst sign-off precedes export. DAG = directed acyclic graph; PII = personal information.

Reference layout: AI Agent Marketplace workbook, Chapter 5, page 18. Technologies and scope: InsightPilot workbook Chapter 5 / architecture D01–D10.